

Zomato Data Analysis Report

Dataset Overview

- The dataset captures restaurant-specific information including location, cuisine offerings, customer ratings, cost metrics, and service-related features.
- The notebook performs a complete exploratory analysis pipeline beginning with data loading and ending with insight generation.

Data Validation

- Checked the dataset for missing entries and duplicate records before analysis.
- Standardized data types for numerical columns to improve computational accuracy.

Analytical Observations

- Computed descriptive statistics for ratings and restaurant pricing.
- Reviewed city-level trends to identify regions with stronger restaurant representation.
- Analyzed cuisine frequencies to determine dominant food preferences.
- Studied customer voting patterns to highlight highly engaged restaurants.

Visualization Summary

- Bar charts were used to compare restaurant counts across cities and cuisine categories.
- Histograms highlighted the overall distribution of customer ratings.
- Trend charts helped assess the relationship between pricing and average ratings.
- Additional plots were used to evaluate the impact of online delivery and table booking services.

Business Insights

- Customer ratings are influenced by multiple operational factors rather than a single variable.
- Restaurant concentration differs considerably across locations, indicating varying market sizes.
- Popular cuisines tend to appear more frequently in cities with higher restaurant density.

Conclusion

- This analysis demonstrates practical applications of Pandas for data manipulation and Matplotlib for visualization.
- The project provides a concise example of transforming raw restaurant data into meaningful business insights through Python.

Notebook code cells analyzed: 13